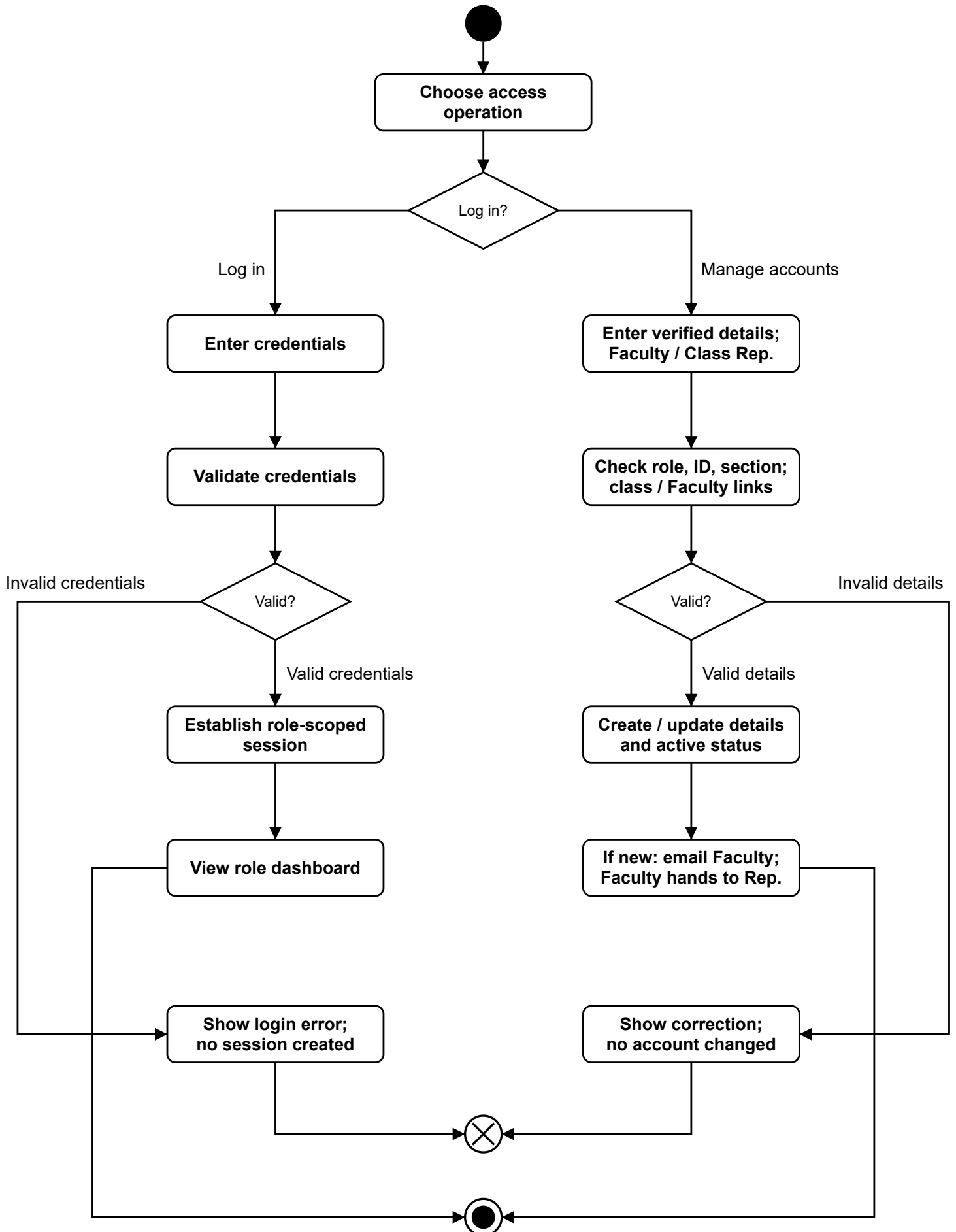
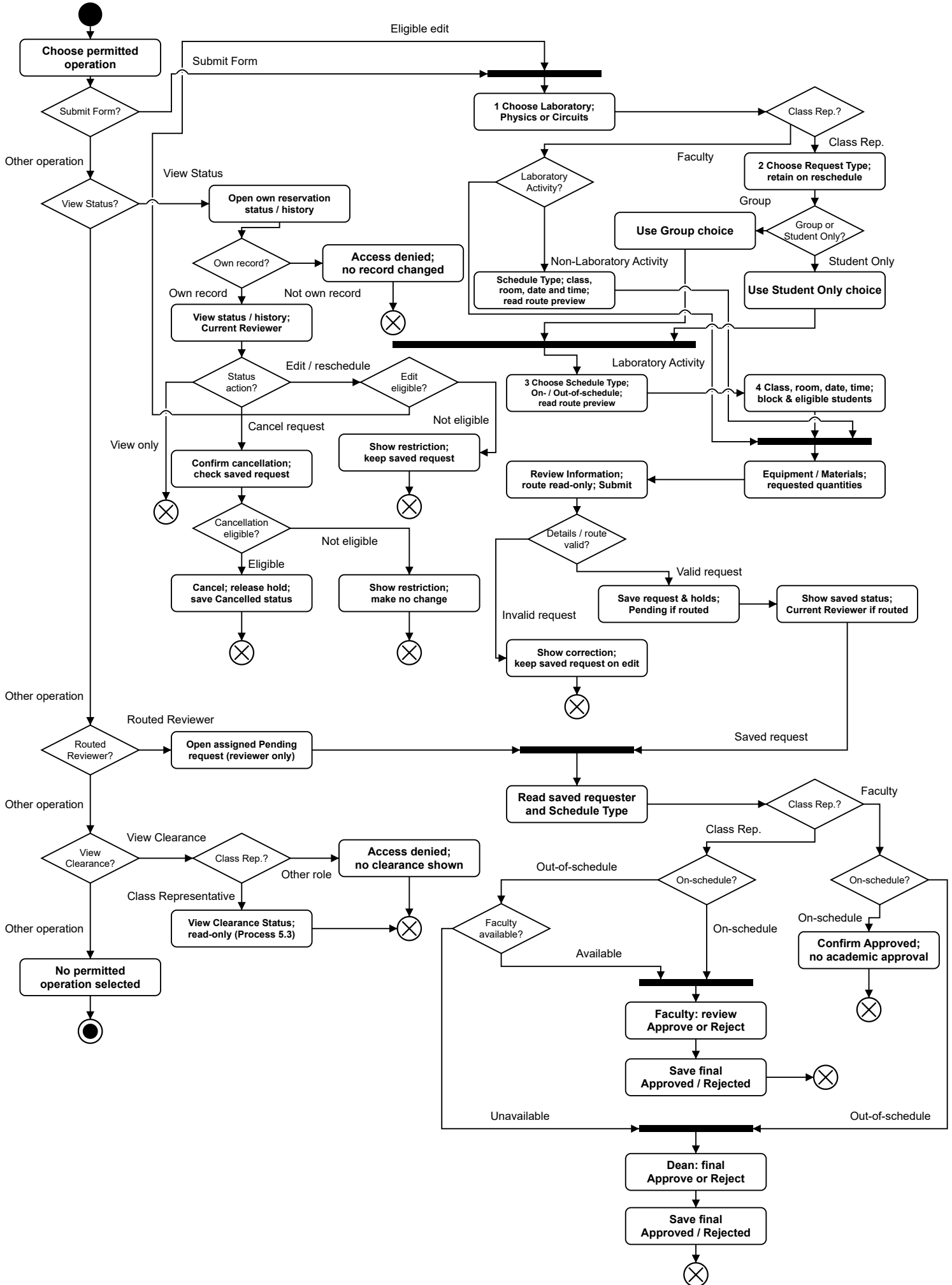


Class Representative	Faculty	Circuit Staff	Physics Staff	Head Lab	Dean
Receive account credentials from Faculty Log in with issued credentials Need lab help? No help needed Help needed Ask lab / equipment info or hours Choose laboratory: Physics or Circuits Group or Student Only? Group Student Only Choose Schedule Type; On / Out-of-schedule; read routing-rule preview Choose class / students; schedule, room and slot Equipment / Materials; requested quantities Review Information; read-only Approval Route; Submit Request Request valid? Invalid Valid Save Pending revision; holds / current reviewer Own Pending / reviewer; Eligible edit / reschedule; revalidate; recheck route; Eligible cancel; release hold Show correction; no new record or hold	Receive Faculty credentials Send verified Class Rep. name, ID and section Receive Class Rep. credentials; hand over Log in with issued credentials Choose laboratory: Physics or Circuits Laboratory Activity? Laboratory Activity Non-Laboratory Choose Schedule Type; class, date and time Choose equipment / materials and quantities Review Information; read-only Approval Route; Submit Request Request valid? Valid Save valid revision; applicable holds / status On-schedule? Out-of-schedule Pending: Dean Own Pending / Dean; Eligible edit / reschedule; revalidate; recheck route; Eligible cancel; release hold Invalid Show correction; no new record or hold On-schedule? Out-of-schedule On-schedule Assigned Faculty available? Available Assigned Faculty: Approve / Reject Faculty approved? Approved Rejected Requester? Class Rep. View own Rejected reservation status Faculty View own Rejected reservation status Requester? Class Rep. View Approved status; selected lab / schedule Faculty View Approved status; booking details Complete; D11 saved Requester? Class Rep. View student, item and clearance status Faculty View own Completed reservation status	Selected lab? Circuits Check approval / stock; prepare Circuits items Issue Circuits items; save Borrowing Slip; status: Ongoing Assigned Staff Selected lab? Circuits Check items; validate; record Returned (good), Broken, Lost, Consumed; save returns / stock All issued qty accounted for?	Physics Check approval / stock; prepare Physics items Issue Physics items; save Borrowing Slip; status: Ongoing Return recorder? Physics Check items; validate; record Returned (good), Broken, Lost, Consumed; save returns / stock No broken / lost Incomplete Keep status Ongoing; remaining qty pending	Create Faculty account; issue credentials Create Class Rep. account and credentials Maintain schedule / daily tasks as needed Review inventory forecast as needed Access Borrowing Slips: Physics and Circuits; issued items / borrowers Head Laboratory Check items; validate; record Returned (good), Broken, Lost, Consumed; save returns / stock; Physics or Circuits Broken or lost? Broken / lost Review outcome / slip; identify accountable student; create clearance View completed logs; export End-Term Report	Review request; Approve / Reject Dean approved? Approved

Login: any role. Create / update Faculty or Class Rep. accounts: signed-in Head Laboratory only.

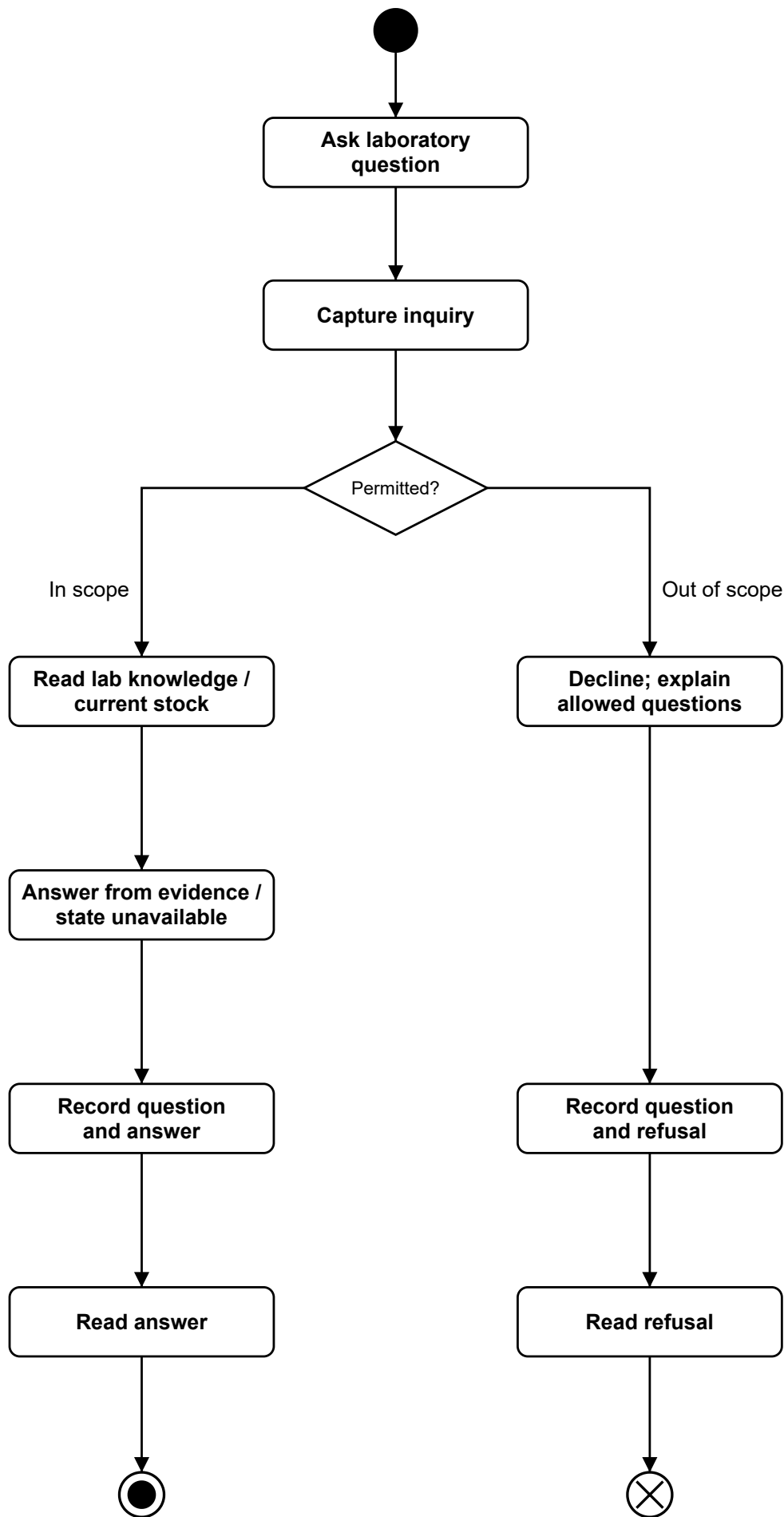


Class Rep.: Group or Student Only for BOTH schedule variants. Student Only = one selected class student.
On-schedule → assigned Faculty. Out-of-schedule → available assigned Faculty, or directly to Dean if unavailable.

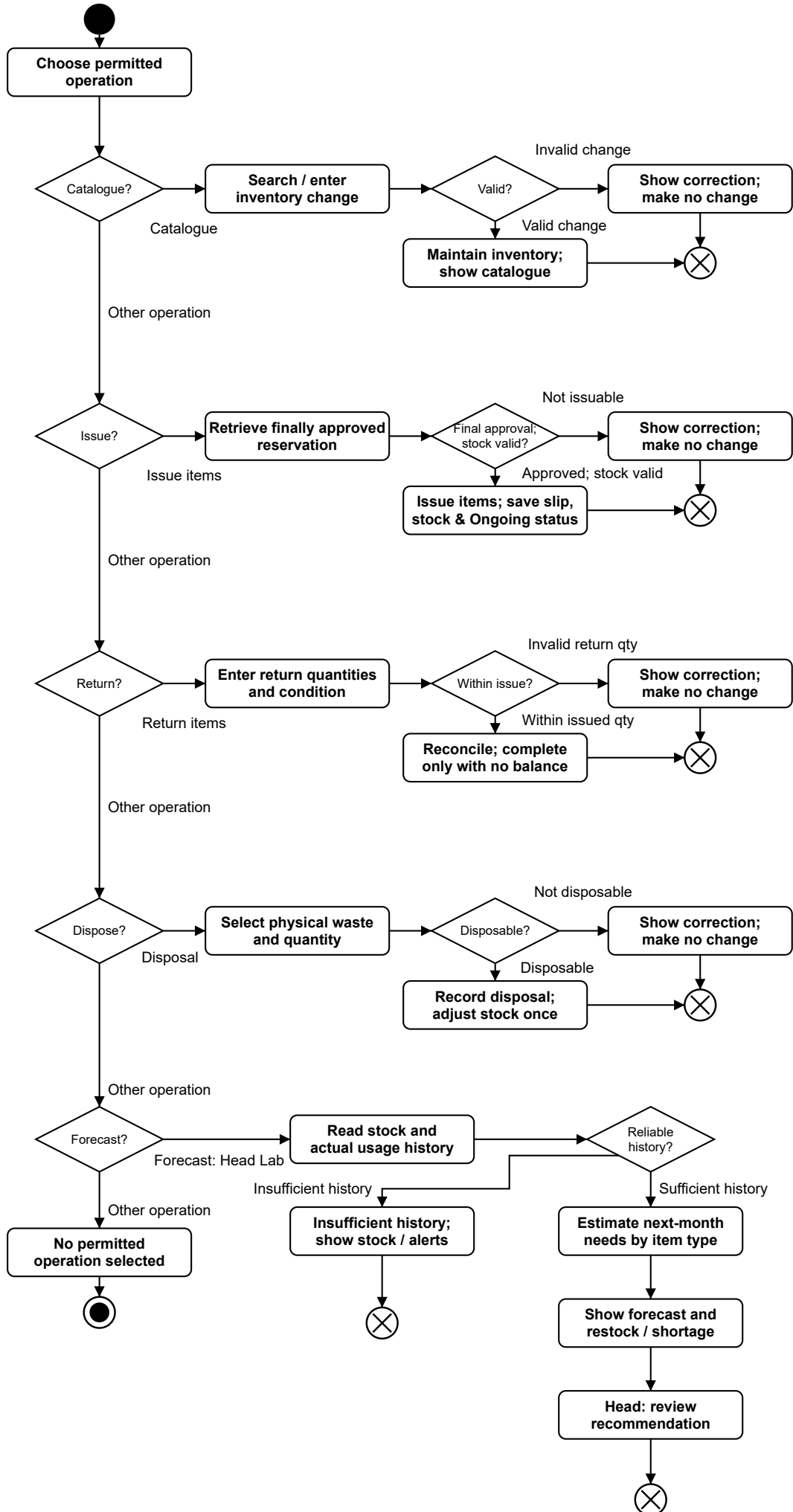


Approved: eligible for Process 4.2. Rejected: release hold. Pending: await reviewer.

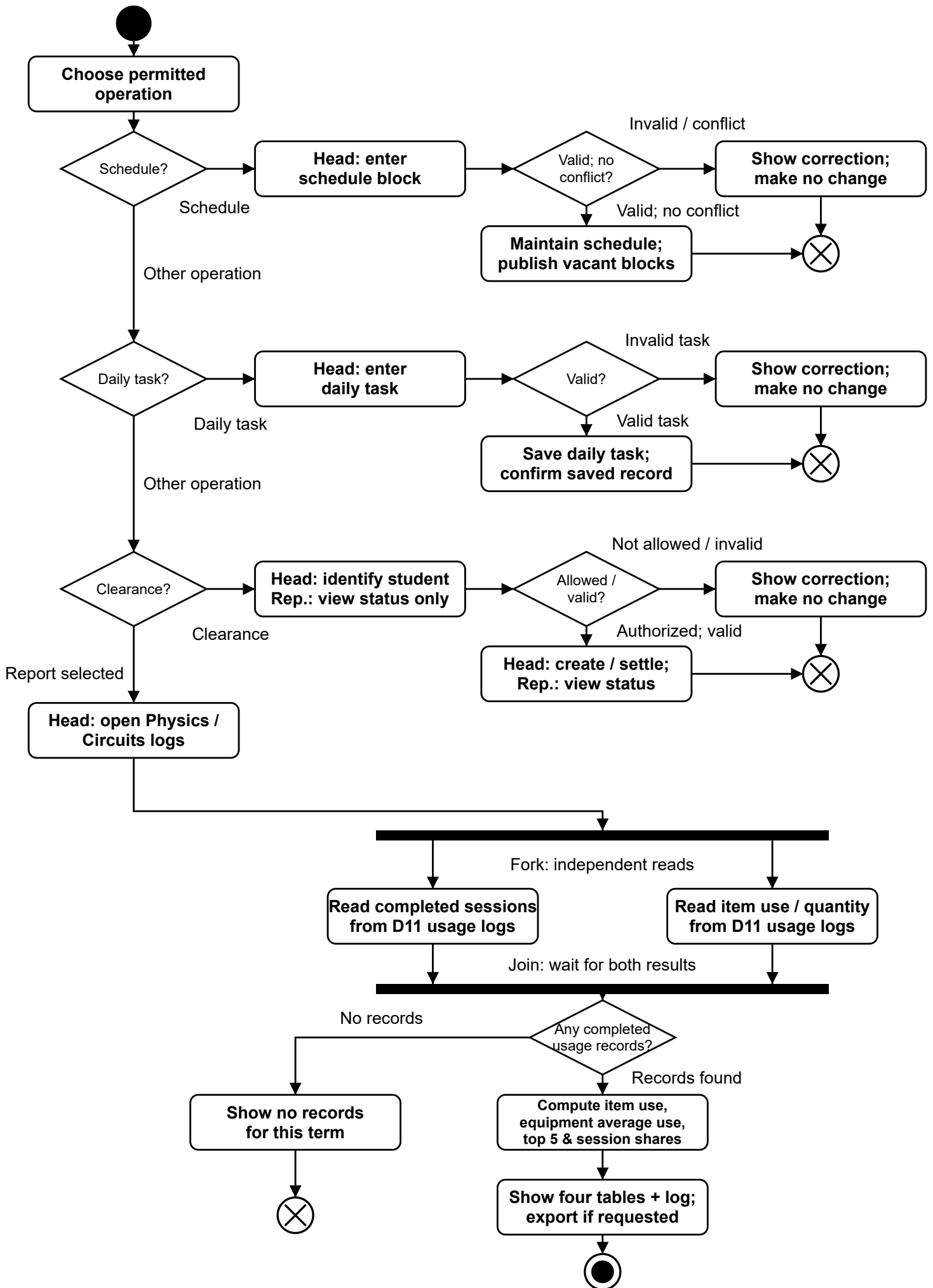
Precondition: signed-in Class Representative or Faculty; informational Q&A only.



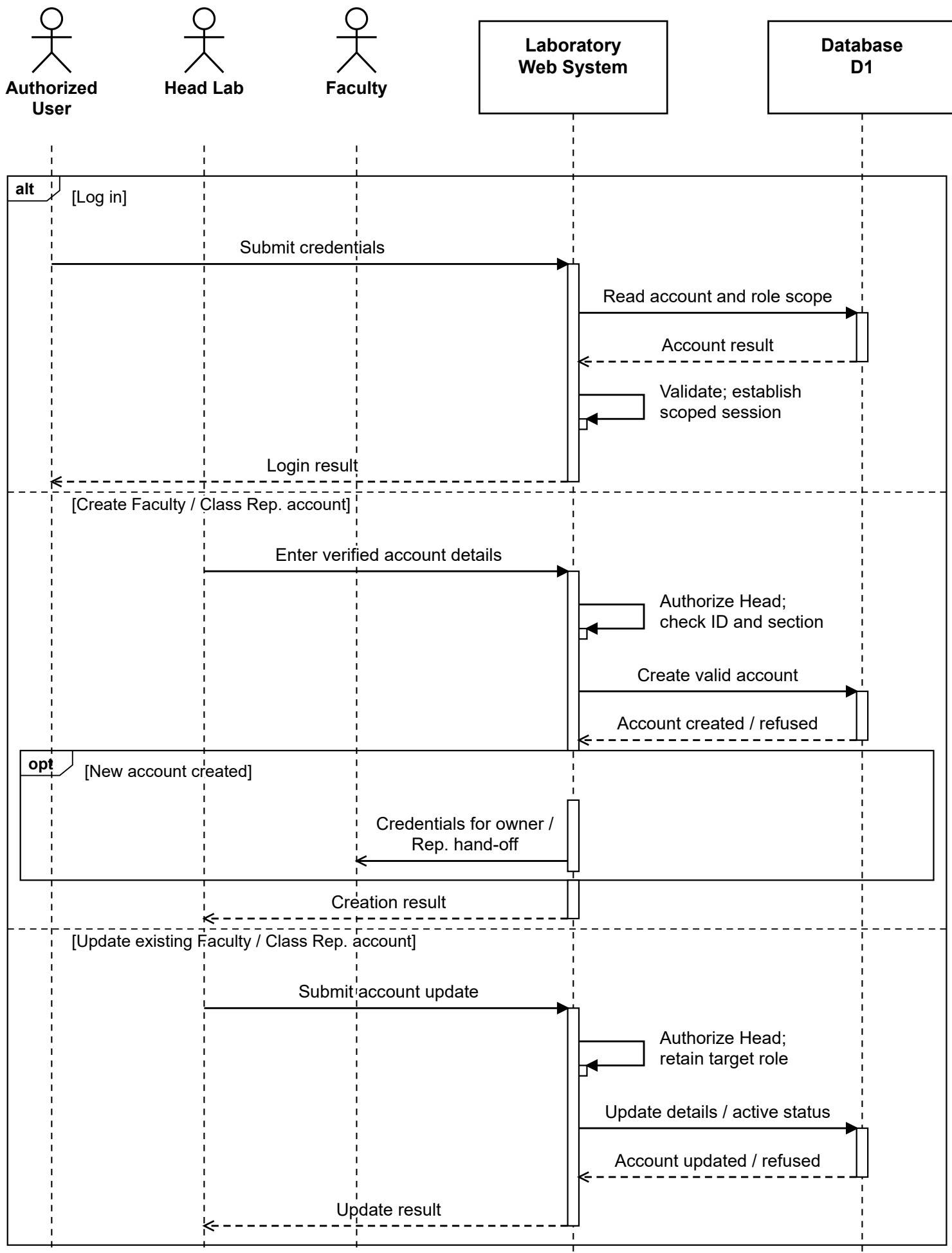
Precondition: signed-in Head Laboratory or Staff within the assigned laboratory.



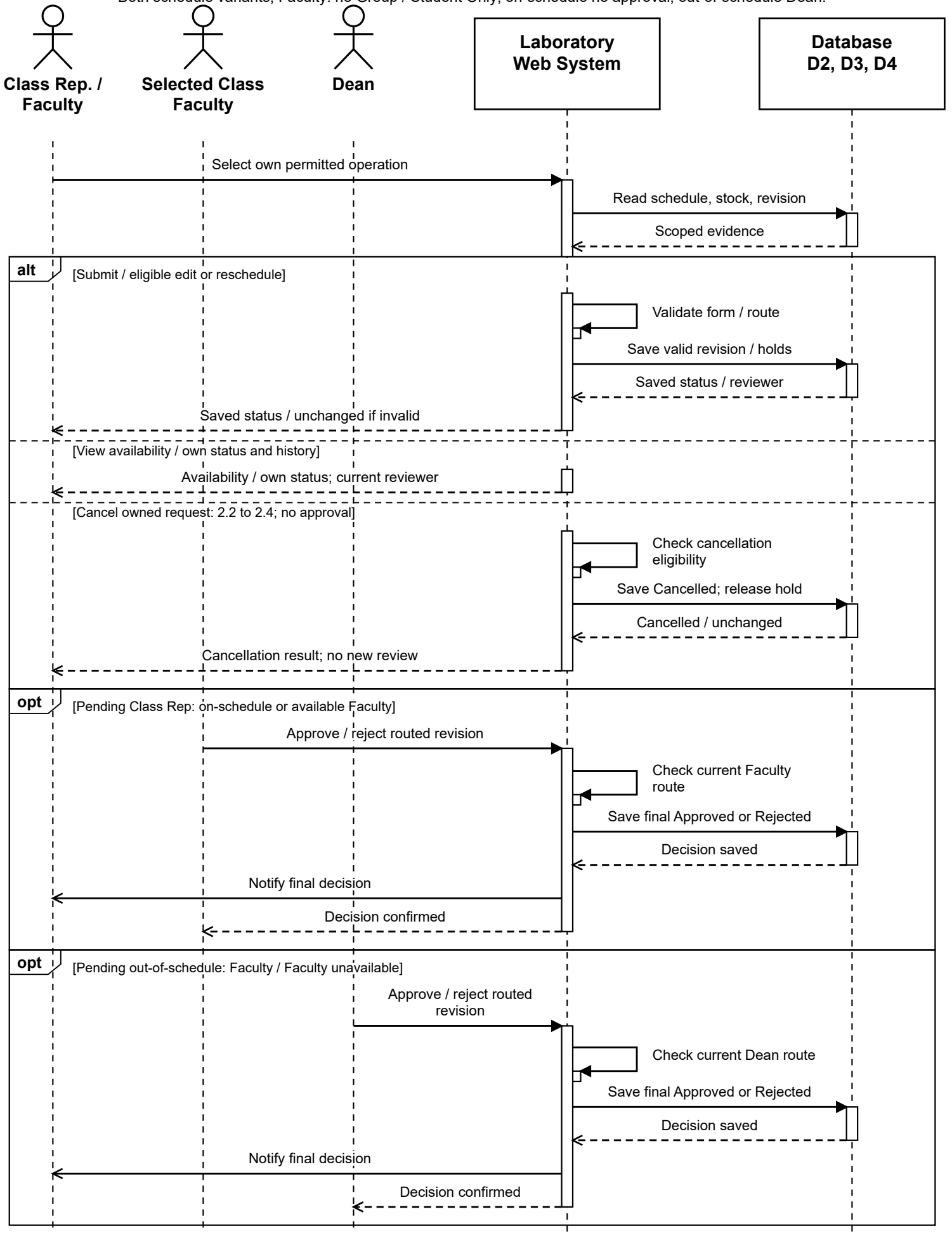
Precondition: signed-in Head Laboratory; Class Representative: assigned-class student clearance only.



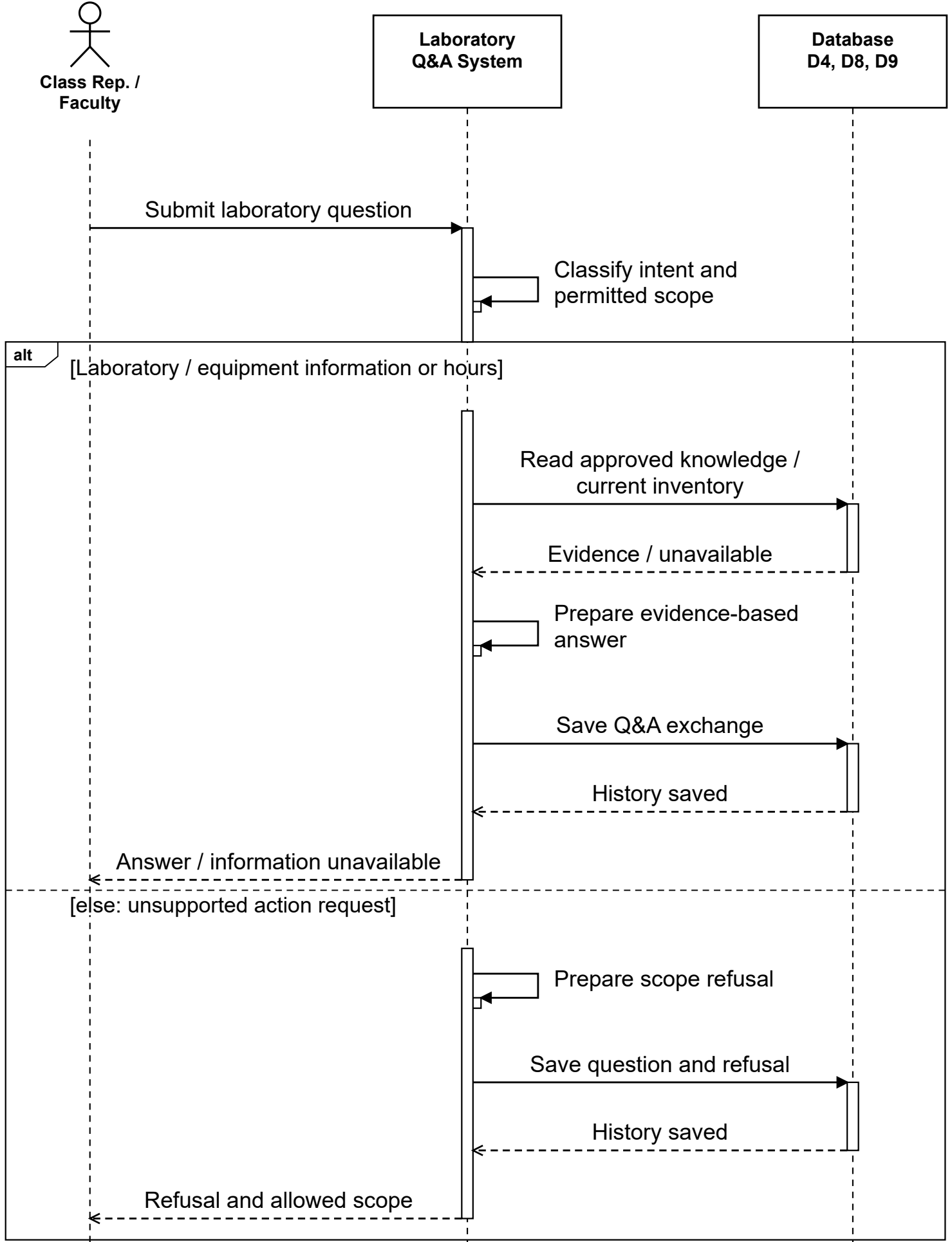
Login uses an issued account. Head creates Faculty first; Faculty manually verifies Class Rep. name, Student ID and section.



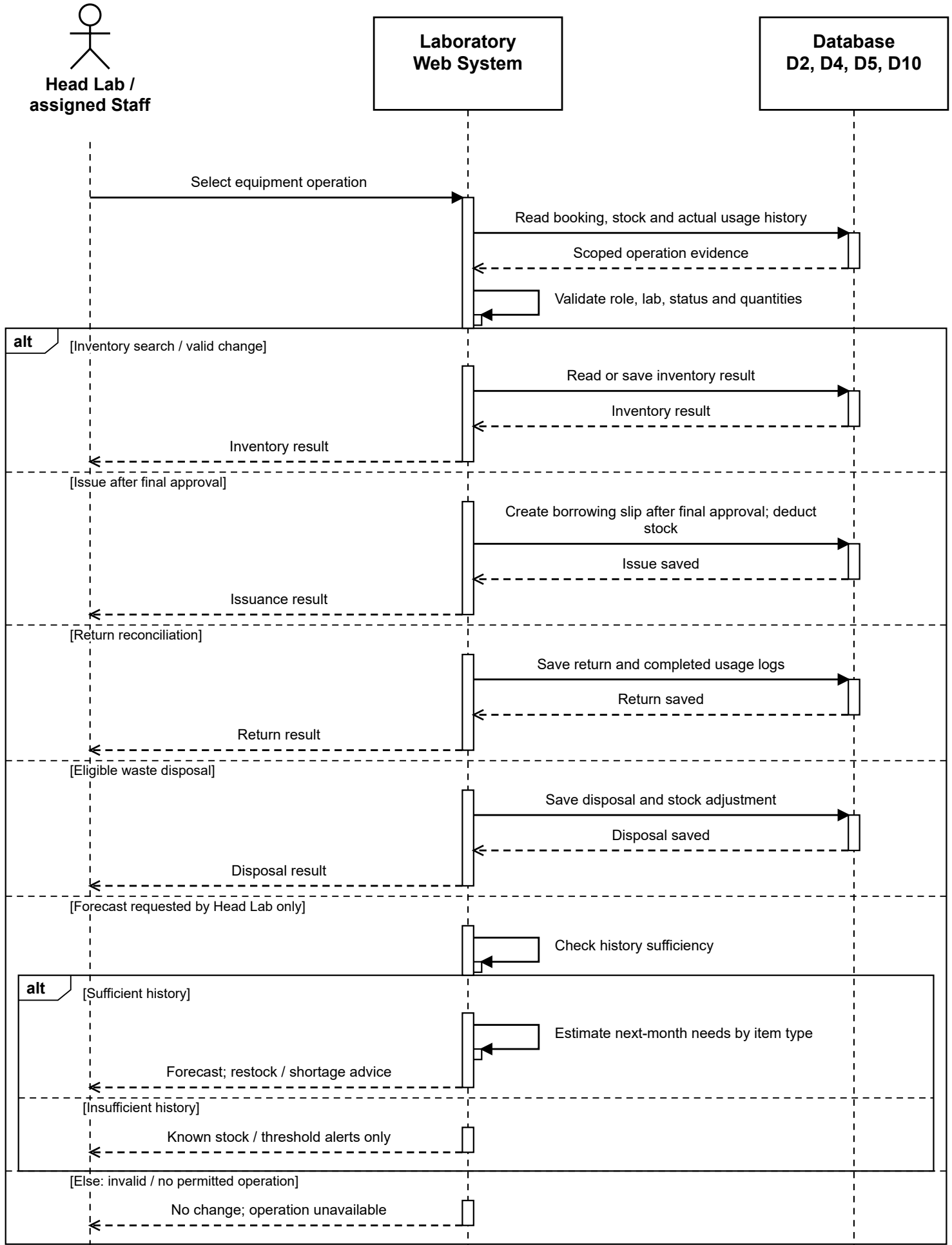
Class Rep.: Group = selected class members; Student Only = one selected class student.
Both schedule variants; Faculty: no Group / Student Only; on-schedule no approval, out-of-schedule Dean.



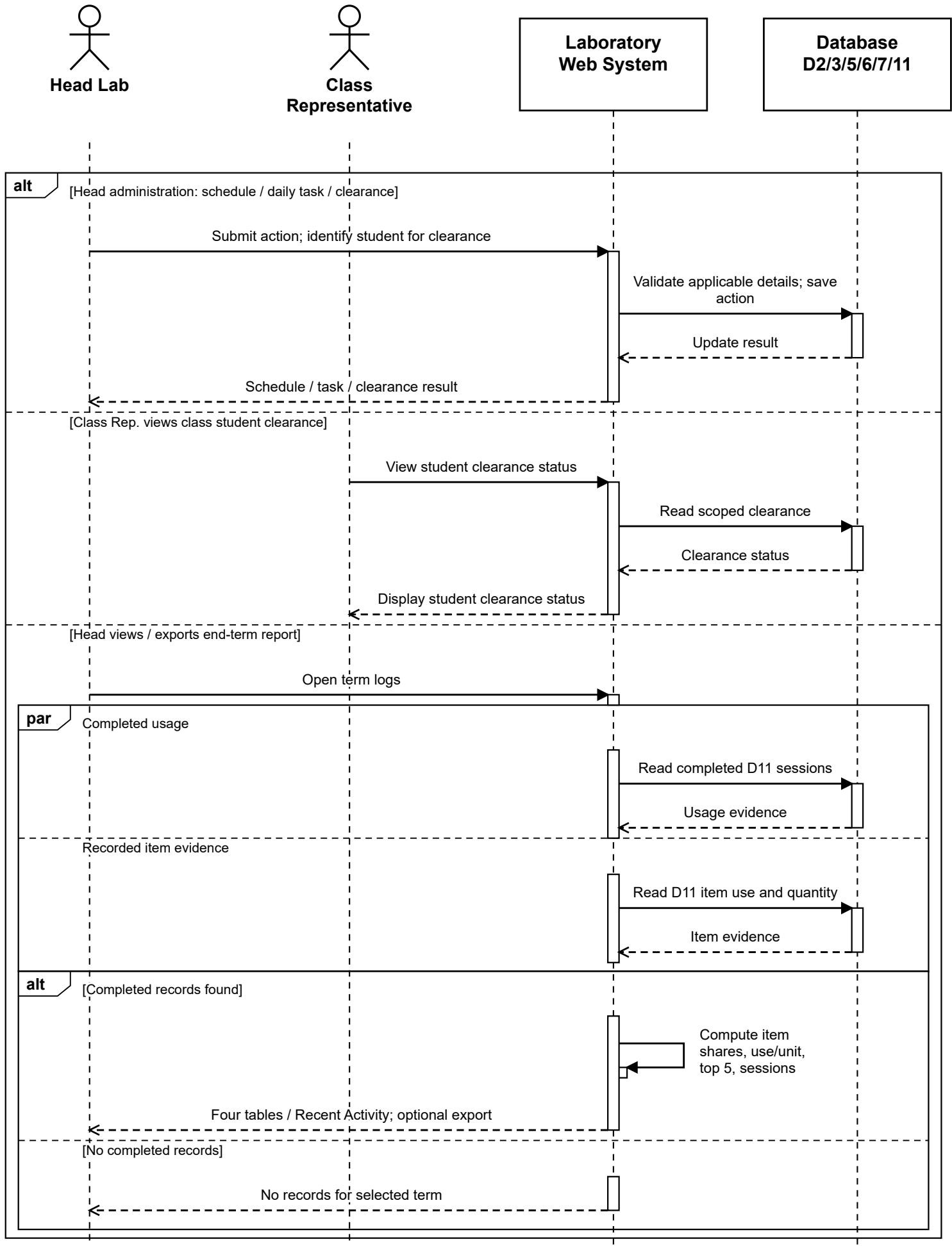
Signed in Class Representative or Faculty; Q&A is informational and scope-limited.



Signed in Head Laboratory or assigned laboratory Staff; laboratory scope is enforced.



Head identifies the responsible student before creating clearance. Class Rep only views own-class clearance status.



Physics and Circuits Laboratory Management System

Deployment Diagram - Proposed architecture

